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Transition state and dynamics of unimolecular no-barrier fragmentation: Thermal dissociation of N₂O₄

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Thermal dissociation reaction into polyatomic molecules on the ground state no-barrier potential energy surface is studied by classical molecular dynamics simulations of N₂O₄ ⇌ 2NO₂. A phase space surface $E_T = V_{\text{eff}}(R_I) \geq 0$ is identified as the transition state (TS), where E_T is the sum of the potential and kinetic energies of interfragment motion and $V_{\text{eff}}(R_I)$ is the orbital angular momentum-dependent effective barrier. By dividing the motion of the system into fragments' vibrational (V), rotational (R), and interfragment (T) modes, where the T mode is composed of translational (TT) and orbital (TL) modes, a scheme of reactive energy transfer for fragmentation is presented. The present energy condition for the TS is in accord with the one of phase space theory (PST). The observed photofragmentation rates of NCNO and CH₂CO which increase with energy slower than predicted by PST suggest that intrareactant energy redistribution may influence the rate. Dissociation is found to occur by energy redistribution among T–R–V modes followed by the one among TT–TL–R modes, which determine the product vibrational and rotational distributions, respectively. This scheme supports separate statistical ensemble method in reproducing the nascent distributions from unimolecular photofragmentation at excess energies above the vibrational threshold. © 1996 American Institute of Physics. [S0021-9606(96)01935-6]

I. INTRODUCTION

In a previous paper, the no-barrier thermal dissociation and association (D/A) reactions N₂O₄ ⇌ 2NO₂ on the ground state potential energy surface in the gas and liquid phases were studied by molecular dynamics (MD) simulations including vibrational degrees of freedom.¹ This system is interesting, in that it is in thermal equilibrium in the gas and liquid phases at room temperatures and that the dissociated molecules have an internal structure beyond that of a diatomic. The rate of dissociation of N₂O₄ by shock waves² and by ultrasonic absorption measurement³ show that the dissociation proceeds according to a Lindemann unimolecular mechanism.⁴ Rate constants for the limiting low concentration and high concentration regions were measured at 253–300 K to be $k_1 = 2 \times 10^{14} \exp(-11\,000/RT)$ dm³ mol^{−1} s^{−1} and $k_\infty = 1 \times 10^{16} \exp(-13\,000/RT)$ s^{−1}, respectively. Applying the Hinshelwood expression for k_1 ,^{5,6}

$$k_1 = Z \frac{1}{(s-1)!} \left(\frac{E_0}{k_B T} \right)^{s-1} e^{-E_0/k_B T}, \quad (1)$$

where Z is the kinetic theory collision number, yields the approximate values for the critical energy $E_0 = 54.2$ kJ/mol and the effective number of oscillators $s = 5$. The rate constant for dissociation of energized N₂O₄ molecules $k_E = k_\infty (k_{-1}/k_1)$ is calculated to be $3.7\text{--}7.8 \times 10^{11}$ s^{−1}, which corresponds to lifetimes $k_E^{-1} = 2.7\text{--}1.3$ ps. In this paper, the dynamics of activated molecules are studied over this time range by classical MD simulations.

To understand the dynamics and the kinetics of unimolecular dissociation, a number of different steps must be studied. Some of the most important are (1) activation of molecules above the dissociation threshold by means of intermolecular energy transfer, (2) deactivation of energized

reactants by intermolecular energy dissipation to surrounding molecules, (3) intramolecular energy redistribution among vibrational, rotational, and reactive modes of highly energized reactants, and (4) the dissociation and energy partitioning among modes of dissociating fragments. Activation and energy relaxation processes (1) and (2) of N₂O₄ in liquid N₂O₄ have been studied in our previous paper.¹ In this paper we focus on steps (3) and (4) of activated molecules in the absence of collisions. We shall study various microscopic reaction processes of N₂O₄ activated above the dissociation threshold in terms of MD simulations. By dividing the total motion of nuclei of the system into fragments' vibrational (V), rotational (R), and interfragment (T) modes, where the reactive mode T is further divided into the interfragment translational mode (TT) and the orbital mode (TL) (see Fig. 1), roles of energy and angular momentum transfers among modes in the D/A reactions will be studied in detail.

Transition state theories (TSTs) of unimolecular dissociation assume that energy is randomized faster than dissociation. However, the experimental and theoretical studies show that intramolecular energy redistribution is not complete for some reaction systems. In the study of overtone initiated dissociation reaction HOOH → 2OH,⁷ quasibiexponential time dependence of product formation suggested some limitations of rapid intramolecular energy randomization hypothesis. Classical trajectory studies of HOOH excited to the 6ν_{OH} level also show evidence for incomplete vibrational energy redistribution within the dissociation lifetimes.⁸ The UV initiated dissociation of HOOH on the directly dissociative potential surface show very fast single exponential rise of product formation ($\tau < 8$ ps), in contrast with the 30–100 ps buildup time for the overtone predissociation. Dissociation rates measured in the threshold region

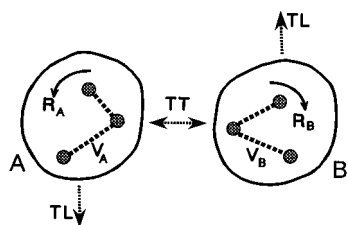


FIG. 1. Degrees of freedom of fragmentation into polyatomic molecules A and B. V and R denote the vibrational and rotational modes of fragments, and TT and TL denote interfragment translational and orbital modes, respectively.

of $\text{HFCO} \rightarrow \text{HF} + \text{CO}$ show strong dependence on the symmetry of the excited quantum states.⁹ These facts suggest that the dissociation rate is not determined simply by the energy E of the microcanonical ensemble, but by the transition probability from strongly mixed vibrational state to states that directly lead to products. This situation clearly violates the hypothesis of rapid energy flow.

The main purpose of this paper is to determine where the real TS in the sense of point of no return to the product is located in phase space, how product state distribution (PSD) is determined, and what kind of adiabaticity exists in the course from reactant to product. We shall derive a unified picture of reactive energy and angular momentum transfer for fragmentation into polyatomic molecules by studying the dynamics of each trajectory initiated from various initial conditions.

This paper is structured as follows. In Sec. II we review statistical theories relevant to the analysis of the simulation data. Methods of the present MD simulation and a model for excitation are presented in Sec. III. The results of MD simulations are analyzed and compared with various TSTs, and a scheme of reactive energy transfer in the D/A reactions is presented in Sec. IV. Finally we summarize our analysis and suggest some future development of the methods discussed in this paper.

II. STATISTICAL THEORIES

In order to understand the fragmentation of activated molecules in the absence of collisions, we first consider the potential energy surface (PES) as a function of a selected reaction coordinate. For reactions without barriers above the energy of products, a TS is neither so clearly defined nor unambiguously located at a specific position along the reaction coordinate.¹⁰ The vibrational degrees of freedom of the parent molecule are conveniently classified into three according to the nomenclature of Refs. 11 and 12: (i) the reaction coordinate R , which is usually defined as the interfragment separation between centers of masses of fragments; (ii) the transitional modes, which change their nature from bending modes in the parent molecule to libration in the bound pair, and to free rotation in the dissociated pair; and (iii) the conserved modes, which retain their vibrational character in both the reactant and products.

Two basic assumptions in deriving the TST rate expression for unimolecular reactions are: (a) the TS is a point of no return to products, and (b) internal energy redistribution is much faster than dissociation. The most used version of TST is the Rice–Ramsperger–Kassel–Marcus (RRKM) theory, which involves a configuration space definition of the TS.⁶ If it is assumed that the TS is located at a particular value of the reaction coordinate, $R = R^\ddagger$, the microcanonical rate constant for unimolecular dissociation at a given energy E and total angular momentum quantum number J is given by

$$k_{EJ} = N_{EJ}^\ddagger / h \rho_{EJ}, \quad (2)$$

where $N_{EJ}^\ddagger$ is the number of available energy levels for modes orthogonal to the reaction coordinate at the TS configuration with energy less than or equal to E , h is Planck's constant, and ρ_{EJ} is the density of states for the reactant at the given E and J . The orthogonal motions include transitional modes as well as conserved vibrations. Many theories with the TS in configuration space are based on Eq. (2), but they differ in the location of the TS.

A. Phase space theory

For bond breaking without barriers, the energy maximum occurs at infinite separation of products. In phase space theory (PST)^{12,13} TS is placed at the barrier R_l of an effective potential $V_{\text{eff}}(R)$, where R_l is calculated as a function of the orbital angular momentum quantum number l of the separating fragments. For a molecule dissociating into two molecular fragments A and B, the angular momentum and energy conservation laws impose the conditions for dissociation,

$$\mathbf{J} = \mathbf{J}_A + \mathbf{J}_B + \mathbf{L}, \quad (3)$$

where $\mathbf{J}$ and $\mathbf{L}$ are the total and interfragment orbital angular momenta, respectively, and

$$E_T = E - E_V - E_R \geq V_{\text{eff}}(R_l), \quad (4)$$

where E_T , E_V , and E_R represent the energies of the T , V , and R modes, respectively. $N_{EJ}^\ddagger$ in Eq. (2) denotes the total number of these vibrational–rotational–orbital states that satisfy the total angular momentum conservation rule and have an energy E_T greater than $V_{\text{eff}}(R_l)$. Because each of these states is equally likely to be formed, PST predicts PSD as well as rates. It is often the case that R_l is so large and the height of the effective barrier is negligible for the largest l 's present. If the l -dependent barriers are completely negligible, PST predicts rates without knowing the PES other than the energy levels of the products.

The PST predictions are in good agreement with the observed no-barrier dissociation rate constants and PSDs near threshold energy in the reactions $\text{NCNO} \rightarrow \text{CN} + \text{NO}$ ¹² and $\text{CH}_2\text{CO} \rightarrow {}^1\text{CH}_2 + \text{CO}$ (where ${}^1\text{CH}_2$ denotes singlet methylene),^{10,14,15} but usually overestimate the rate constants at high energies. PST underestimates the vibrational excitation and overestimates the rotational and translational excitations when product vibrations are energetically accessible.^{14–16} The variational RRKM theory and the statis-

tical adiabatic channel model are the models that incorporate dynamical constraint that slows down the reaction at high energies.

B. Variational RRKM theory

Variational RRKM theory predicts rates by assuming that any point $R=R^\ddagger$ on the reaction coordinate can be a TS and the best TS is placed at the value $R^\ddagger$, which minimize the flux, i.e., where $(\partial N_{EJ}^\ddagger/\partial R)_{R=R^\ddagger} = 0$.^{11,17} An important consequence was that the TS position moved in along the reaction coordinate as the energy increased. In the new version of variational RRKM, which assumes adiabaticity of the conserved modes,¹⁷ the dissociation rate on each vibrationally adiabatic (VA) surface i is calculated using Eq. (2) and the number of accessible states for the transitional modes at $R=R_i^\ddagger$,

$$N_{iEJ}^\ddagger = \int_0^{E-E_i^\ddagger} \rho_J(\epsilon) d\epsilon, \quad (5)$$

where $\rho_J(\epsilon)$ is the density of states for the transitional modes at a given energy ϵ and $E_i^\ddagger$ is the potential energy for the i th vibrational state. The variationally determined TS at $R_i^\ddagger$ is considered to be looser as the vibrational energy of i increases. The transitional modes and the reactive mode are assumed to continue their free interchange of energy until the interfragment separation reaches R_l . The dissociation rate is given by the sum of rates calculated by Eqs. (2) and (5) on each VA surface, and the vibrational state distribution is given by the ratios of these rates. The variational RRKM method, in combination with the assumption of adiabatic vibrational and statistical rotational motions, was found to give a good description of photodissociation rate constants and PSDs of NCNO¹² and CH₂CO.^{14,15} However, a curious difference was pointed out as to the effect of PES to the rates and PSD.¹² Rates are thought to be determined by a hypothetical surface that minimizes the flux, while PSD is determined at R_l that is generally farther toward products than the variationally determined TS.

C. Statistical adiabatic channel model

In the statistical adiabatic channel model (SACM) developed by Quack and Troe,¹⁸ each channel, which connects a reactant quantum state to a particular product state, has its own effective energy as a function of a reaction coordinate. Fewer channels are accessible in this model than in PST because a channel is closed if the maximum in its effective energy curve lies above the total energy. Although this theory gave an energy dependence of the dissociation rate of CH₂CO better than PST by raising the product state threshold, experimental data showed that all states were observed precisely at their thermochemical threshold.¹⁵

D. Separate statistical ensembles

To explain PST-like rotational state distributions but higher vibrational excitations of products, Witting *et al.* proposed a modified version of PST, which is called the separate

statistical ensembles (SSE) method.¹⁶ In SSE, the product vibrational state distribution is calculated first from an ensemble of vibrational degrees of freedom of the parent molecule, and second, the product rotational and translational excitations are estimated for each product vibrational state from an ensemble of transitional modes, reactive mode, and parent rotational modes. The calculated PSDs reproduced much more complicated variational RRKM calculations, and gave excellent agreement with the nascent distributions from unimolecular dissociation of monoenergetic NCNO¹⁶ and CH₂CO.¹⁵

E. TS based on interfragment bottleneck in phase space

Vibrational predissociation rates of van der Waals molecules XI₂ (X=He, Ar)



where v denotes the vibrational quantum number in the I₂ part, show strong dependence on the specific energy level prepared by stimulated emission pumping. The standard RRKM theory assumption that unimolecular complexes exist for $R < R^\ddagger$ was shown not valid, but a statistical rate that assumes a TS in phase space was found to describe some aspects of fragmentation of van der Waals molecules.^{19,20} The TS was represented by a phase space surface,

$$\frac{P^2}{2\mu} + V_T(R) = 0, \quad (7)$$

where P and R are the canonical momentum and coordinate associated with the van der Waals degrees of freedom, μ is the effective mass, and V_T is the effective interaction potential.

Rates calculated by quasiclassical trajectory simulations initiated from excited states (v, n) , which correspond to vibrational action v in the I₂ part and n in the van der Waals degree of freedom, and the ones calculated by quantum mechanics based on the Fermi golden rule approach were both very nonstatistical; they did not increase monotonically, but oscillated with energy. The rate constants obtained based on the phase space bottleneck as TS [Eq. (7)] agreed reasonably well with the classical and quantum rate constants averaged over the van der Waals manifold of states, which was a remarkable improvement over standard RRKM theory.

Two kinds of statistical theories of microcanonical rates for unimolecular dissociation; one is based on the TS in configuration space and the other on the TS in phase space, have been reviewed. As we have seen above, there are many situations that clearly violate the hypothesis of rapid intrareactant energy flow, and hence in which energy randomization among degrees of freedom influences the rate. In this paper, various transition state theories that place the TS in configuration space and more generally in phase space will be examined from the dynamical point of view.

TABLE I. Moments of inertia and force constants of NO₂ in the internal coordinate representation.

Moment of inertia	(10 ⁻⁴⁶ kg m ²)	Potential energy term	Force constant ^a
I_x	6.441 21	f_{11}, f_{22}	9.5749
I_y	0.329 86	f_{12}	1.1860
I_z	6.771 07	f_{33}	1.7385

^aUnits—energy is in 10⁻¹⁸ J, and coordinates are in Å and rad.

III. MOLECULAR DYNAMICS SIMULATION

As a preliminary step of this paper, we carried out non-equilibrium MD simulations initiated from various excited states of a reactant NO₂ pair in 124 N₂O₄ molecules as a solvent, and the details of the simulation were described in Ref. 1. In this paper we focus on the dynamics in the gas phase of a reactant NO₂ pair activated above the dissociation threshold on the ground state potential energy surface. We shall study the roles of energy and angular momentum transfer among various modes in the D/A reactions.

In our study of the dissociation dynamics, the 12 vibrational degrees of freedom of parent N₂O₄ molecule are conveniently classified as in the last section into three: one reactive mode (ν_3), five transitional modes ($\nu_4, \nu_6, \nu_7, \nu_8, \nu_{10}$), and six conserved modes ($\nu_1, \nu_2, \nu_5, \nu_9, \nu_{11}, \nu_{12}$) in the nomenclature by Herzberg.^{21,22} A detailed *ab initio* potential energy surface for the present system is not available, and a model potential for a loosely interacting pair of NO₂ molecules is employed. The internal coordinates of NO₂ are $S_1 = \Delta r_1$, $S_2 = \Delta r_2$, $S_3 = \Delta \phi_{12}$, where r_1 and r_2 are the bond lengths between N and O, and ϕ_{12} is the angle between bonds 1 and 2. Intrafragment potential energy V_{intra} for the conserved modes is assumed to be harmonic, and the force constants in the internal coordinate representation and the moments of inertia are given in Table I.

The intermolecular potential V_{inter} between two NO₂ is taken to be the sum of a Morse potential for the NN bond with the dissociation energy $D_e = 56.9$ kJ/mol $= 0.0945 \times 10^{-18}$ J/molecule, and Lennard-Jones potentials between N–O and O–O atoms of parameters given in Table II. Lennard-Jones potentials between N–O are added anew to the potential used in our previous paper,¹ but this revision gives little difference on the interfragment potential energy and the equilibrium NN distance. We name this revised potential Morse+LJ as in Ref. 1. All atoms interact through V_{intra} and V_{inter} , and the force $\mathbf{F}_i$ on the i th atom is given by

$\mathbf{F}_i = \nabla_i V_{\text{intra}} + \nabla_i V_{\text{inter}}$. The full coupling among three classes of modes, as well as with parent rotation, is included in the present simulation.

Starting from a randomly chosen time point in the file of MD of equilibrium liquid N₂O₄ (125 N₂O₄) at 293 K, where all N₂O₄ evolve under intramolecular harmonic potential,²² a time was chosen when the oscillating NN distance R_{NN} of a selected N₂O₄ reached its minimum distance (R^s), and drew two NO₂ fragments in the direction of NN until R_{NN} became a selected value R^{ex} , keeping the center of NN unchanged. The simulation of the activated reactant was started from this selected initial configuration by recognizing N₂O₄ as two NO₂ molecules interacting through the Morse+LJ potential, keeping the momenta unchanged. If we denote the Morse+LJ potential energies at R^s and R^{ex} as E^s and E^{ex} , respectively, the interfragment mode is given an energy $E^{\text{ex}} - E^s$ by the selected initial condition, and the excess energy was found to distribute quickly to intrafragment vibrational modes.¹ The time step (Δt) was set to 1/2048 ps, and the simulation was carried out both forward and backward in time for 128 ps until the reactant dissociates.

From each trajectory of a pair of NO₂ molecules, we stored as a function time: (1) R the distance between centers of masses of two NO₂ molecules, where $R = R_{\text{NN}} + 0.63$ Å in the collinear configuration; (2) V_{intra} the sum of internal potential energies for the stretching and bending modes of two NO₂; (3) V_{inter} the NO₂–NO₂ intermolecular potential energy; (4) $E_{\text{kin}}^{\text{trans}} = E_{\text{kin}}^{\text{c.m.}} + E_{\text{kin}}^{\text{rel}}$, where $E_{\text{kin}}^{\text{c.m.}}$ and $E_{\text{kin}}^{\text{rel}}$ represent the kinetic energy of the center of mass of the NO₂ fragments as a whole and the one of the relative motion between the centers of masses of fragments, respectively; (5) E_R , the sum of the rotational kinetic energies of two NO₂; (6) $E_V = V_{\text{intra}} + E_{\text{kin}} - E_{\text{kin}}^{\text{trans}} - E_R$, the vibrational energy of NO₂ molecules, where E_{kin} is the total kinetic energy of the system; (7) $E_T = E_{\text{kin}}^{\text{rel}} + V_{\text{inter}}$, the sum of kinetic and potential energies of the interfragment motion; (8) $E = V_{\text{inter}} + V_{\text{intra}} + E_{\text{kin}} - E_{\text{kin}}^{\text{c.m.}} = E_T + E_R + E_V$, the total energy in the center of mass scheme; and (9) $\mathbf{J}^2/\hbar^2 = J(J+1)$ and $\mathbf{L}^2/\hbar^2 = l(l+1)$, where $\mathbf{J}$ and $\mathbf{L}$ are the total angular momentum and interfragment orbital angular momentum, respectively. In point (4), translational kinetic energy is evaluated according to the following formulas:

$$E_{\text{kin}}^{\text{trans}} = \frac{1}{2} m \dot{\mathbf{r}}_{G1}^2 + \frac{1}{2} m \dot{\mathbf{r}}_{G2}^2, \quad (8)$$

$$E_{\text{kin}}^{\text{c.m.}} = \frac{1}{2} (2m) \left(\frac{\dot{\mathbf{r}}_{G1} + \dot{\mathbf{r}}_{G2}}{2} \right)^2, \quad (9)$$

$$E_{\text{kin}}^{\text{rel}} = \frac{1}{2} \mu \dot{\mathbf{R}}^2, \quad (10)$$

where $\mathbf{r}_{G1}$ and $\mathbf{r}_{G2}$ are the position vectors of the centers of masses of NO₂ molecules of mass m , $\mathbf{R} = \mathbf{r}_{G2} - \mathbf{r}_{G1}$, and $\mu = m/2$ denotes the reduced mass. The kinetic energy of relative motion between fragments is composed of contributions from relative translation E_{kin}^t and orbital motion E_{TL} as

$$E_{\text{kin}}^{\text{rel}} = E_{\text{kin}}^t + E_{\text{TL}}, \quad (11)$$

$$= \frac{P^2}{2\mu} + \frac{\mathbf{L}^2}{2\mu R^2}, \quad (12)$$

TABLE II. Atom–atom potential parameters for the NO₂–NO₂ intermolecular interaction used in the MD simulations.

Interaction	Type	Parameters		
N–N	Morse	$D_e (10^{-18} \text{ J})$	$\alpha (\text{\AA}^{-1})$	$r_0 (\text{\AA})$
		0.0945	2.4987	1.782
N–O	Lennard-Jones	$A (10^{-18} \text{ J \AA}^6)$	$B (10^{-18} \text{ J \AA}^{12})$	
		0.5978	115.14	
O–O	Lennard-Jones	1.4177	646.3	

where $P = \mu \dot{R}$ denotes momentum of relative translation and $L = R \times P$ denotes the angular momentum of interfragment orbital motion with moment of inertia μR^2 . The reactive mode energy E_T is hence composed of contributions $E_{TT} = E_{\text{kin}}^T + V_{\text{inter}}$ from the interfragment translational mode and E_{TL} from the orbital mode as

$$E_T = E_{TT} + E_{TL}, \quad (13)$$

which is schematically shown in Fig. 1. The effective potential V_{eff} for relative translation ($E_T = E_{\text{kin}}^T + V_{\text{eff}}$), when the interfragment orbital angular momentum L is conserved, i.e., $L^2 = \hbar^2 l(l+1)$, is given as

$$V_{\text{eff}} = V_{\text{inter}} + \frac{\hbar^2 l(l+1)}{2\mu R^2}. \quad (14)$$

An l -dependent barrier is formed at $R = R_l$, where there is a maximum in $V_{\text{eff}}(R)$, and the barrier becomes tighter (R_l decreases) and higher [$V_{\text{eff}}(R_l)$ increases] as l increases. In point (5), E_R is evaluated by using the moments of inertia of NO_2 given in Table I.

IV. RESULTS AND DISCUSSION

A. Analysis of results

We carried out about 2000 MD simulations in the gas phase for reactants excited to various degrees ($E = 0 \sim 2.2 \times 10^{-18}$ J) from various initial conditions, and analyzed the time evolution of energies and angular momenta of each trajectory in detail. Figures 2, 4, 5, 7–9 illustrate examples of trajectories of reactant NO_2 pairs in the gas phase: (a) total energy E in the center of mass scheme and energy partitioning among E_T , E_R , and E_V ; (b) the time-dependent interfragment distance R between centers of masses of NO_2 molecules; (c) E_T , E_{kin}^T , and V_{eff} during the approaching and separating processes of fragments; and (d) $J(J+1)$ and $l(l+1)$.

Total energy E and total angular momentum $J(J+1)$ are conserved in the absence of collisions. Time zero is when two NO_2 in a compressed N_2O_4 configuration were released at a selected R^{ex} value with zero kinetic energy of relative translation. All trajectories initiated from activated configurations in the gas phase can be classified into four types: (i) DD, the direct scattering trajectory; (ii) DAD, where a pair collides, remains associated for some time (lifetime τ) and then dissociates; (iii) DA or AD, where a pair remains associated forward or backward in time during the simulation time over 128 ps; (iv) AA, where a pair is associated during the simulation time of 256 ps. All trajectories of type (iii) and some of type (iv) will be DAD if the MD simulation is continued for much longer time.

Figure 2 shows a trajectory initiated at $R^{\text{ex}} = 1.460 \text{ \AA}$ ($E = 0.094 \times 10^{-18}$ J). A reactant pair collides and associates at an inner turning point (ITP) B by TV energy transfer from the T to the V mode. Energy redistribution occurs among $T-R-V$ modes while the pair remains associated, and then dissociates at F by VT (from V to T) energy transfer. This trajectory is a DAD type, and E_T changes sign (+, −, +) accordingly. The lifetime τ of the A state of this trajectory is

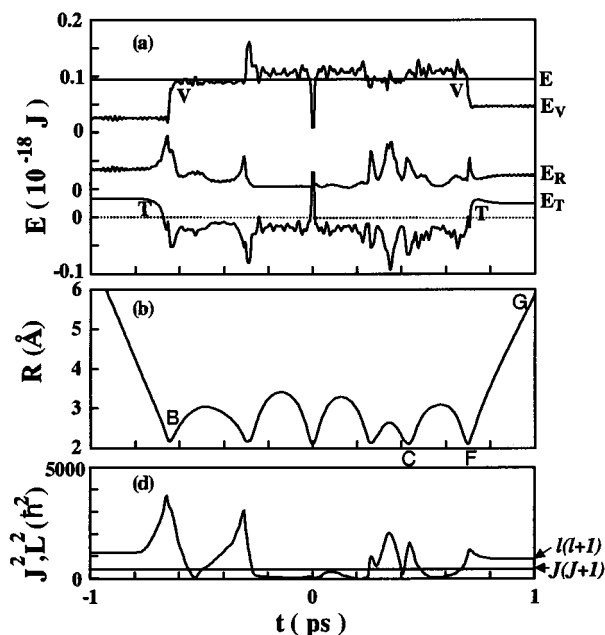


FIG. 2. Time-dependent partitioning of energy and angular momenta of a reactant NO_2 pair in the gas phase at $E = 0.094 \times 10^{-18}$ J: (a) total energy E and energy partitioning among E_T , E_R , and E_V , where three ordinates are for E and E_V , E_R , and E_T from top to bottom; (b) interfragment distance R between centers of masses of NO_2 molecules; (c) total angular momentum $J(J+1)$ and orbital angular momentum $l(l+1)$.

1.4 ps. Energy transfers due to the coupled motion of R and T modes occur around the TV and VT energy transfers at B and F, but they do not play dominant roles in the D/A reactions, i.e., in changing the sign of E_T . However, these energy fluctuations affect the prior rotational and translational energy distributions leading to association and product state distributions after fragmentation. The fragments approach or depart at positive constant E_T in the D state, and E_V and E_R are also kept constant. This shows that the vibrational and rotational motions are those of free molecules in the D states, irrespective of the interfragment separation R . This observation suggests an inability to calculate the number of states as a function of R according to Eq. (5). When E_T is large enough, the motion along the reaction coordinate may be as rapid as, or more rapid than, the librational motion of the transitional modes, and Eq. (5) for the number of states in variational RRKM theory will not be applicable. The number of accessible states seems to be more close to the one for the free rotor asymptote considered in PST.

Interfragment orbital angular momentum $l(l+1)$ fluctuates periodically during the pair is associated, but it is kept constant in the D states. Its fluctuation is in accord with the one of kinetic energy of orbital motion E_{TL} because they differ only by the factor $\hbar^2/2\mu R^2$. Orbital angular momentum $l(l+1)$ larger than $J(J+1)$ is countered by the angular momenta of fragments, thereby conserving J . A large fluctuation in $l(l+1)$ is observed upon dissociation since the motion along the reaction coordinate is as rapid as bending, and the bending motion takes on a half-collision character. This effective anisotropy in the bending potential couples orbital

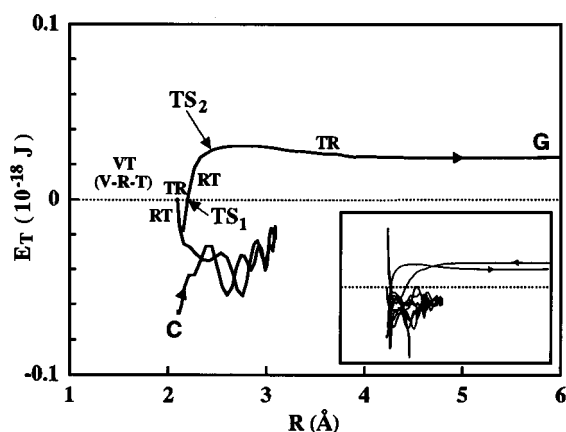


FIG. 3. Reactive mode energy E_T versus interfragment separation R of the dissociation process [from C to G in Fig. 2(b)] and the one of the whole trajectory in the inset. A point of no return of the trajectory is at TS_1 , where $E_T=0$, or TS_2 , where the T mode gets enough energy to dissociate. Energy transfer responsible for each energy fluctuation in E_T is shown in the figure.

and product angular momenta, and the PSD is determined finally through this coupling. The l -dependent barrier in this trajectory is low and is at large separation $R_l=6$ Å because of low orbital angular momentum $l(l+1)\approx 860$.

Figure 3 plots reactive mode energy E_T versus interfragment separation R of the dissociation process [from C to G in Fig. 2(b)] and the one of the whole trajectory in the inset. A point of no return of the trajectory is at TS_1 , where $E_T=0$, or TS_2 , where the T mode gets enough energy to dissociate. Both candidates of the TS are located at the inner side of a considerable part of the trajectory in the associated state. This shows a difficulty of defining the TS in coordinate space alone, even if the total energy dependence is included; as in RRKM theory. The reason for each energy fluctuation of E_T is shown in the figure. E_T goes up by RT energy transfer, down by TR , then RT and TR follow to dissociation. The main VT process, which occurs around an ITP F, is found to occur directly and via the $V-R-T$ path.

Figure 4 also shows a DAD trajectory at $E=0.127\times 10^{-18}$ J. A pair collides and associates by TR energy transfer around an ITP B, intermode energy redistribution continues for 7.7 ps during the pair is associated, TR followed by RT energy transfer, which occurs around an ITP E_1 , terminates at an outer turning point (OTP) E_2 , where E_T changes sign and the pair dissociates. The T and R modes are strongly coupled in the A state, and in-phase energy evolution of opposite sign with the period half of the one of interfragment motion is seen in E_T and E_R , while significant vibrational energy transfer is much less often. It is noted that E_V from C_1 to C_2 in the A state is lower than E_V in the D states. This shows that the rotational energy is not always available to the reactive mode, and the assumption of adiabatic vibrational and statistical rotational motions used in variational RRKM theory is not justified here microdynamically.

After E_T has acquired positive energy by VT and/or TR followed by RT energy transfer, E_T is kept positive constant

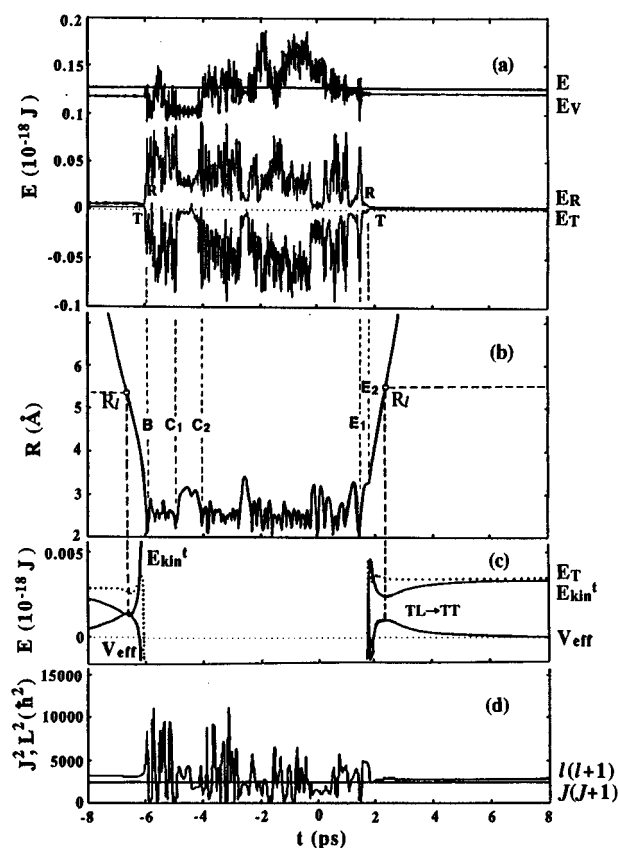


FIG. 4. Time-dependent partitioning of energy and angular momenta of a reactant NO_2 pair in the gas phase at $E=0.127\times 10^{-18}$ J: (a) total energy E and energy partitioning among E_T , E_R , and E_V ; (b) interfragment distance R ; (c) E_T (dotted line), E_{kin}^t and V_{eff} (full lines) during the approaching and separating processes of fragments; (d) $J(J+1)$ and $l(l+1)$. The position of the orbital angular momentum dependent effective barrier R_l is shown in the figure.

and no more energy transfer among $T-R-V$ modes is observed in the D state. Kinetic energy of interfragment orbital motion E_{TL} is transferred to interfragment translational energy E_{TT} during the former decreases proportionally to R^{-2} with almost constant $l(l+1)\approx 2900$, and as a result, E_{kin}^t becomes minimum at an l -dependent potential barrier R_l of V_{eff} [Fig. 4(c)].

Figure 5 shows a trajectory initiated at $R^{\text{ex}}=1.5051$ Å ($E=0.078\times 10^{-18}$ J). Association occurs at B by TV energy transfer. Energy fluctuation due to the coupled motion of T and R does not play a dominant role in changing the sign of E_T at B. Energy E_T changes sign from negative to slightly positive at C_2 by VT and RT energy transfer, the fragments depart to a maximum $R=4.57$ Å at E, unable to dissociate, then come back to F, where the T mode gains enough energy to dissociate by RT energy transfer. The trajectory is reflected at E with positive E_T because the T mode does not have enough energy to pass over the l -dependent potential barrier of V_{eff} , which is slightly positive, as shown in Fig. 5(c). In other words, the pair cannot dissociate because $E_{TT}<0$ from C_2 to F. Coupled motion of the V and R modes disappears at F, and no more intermode energy transfer is

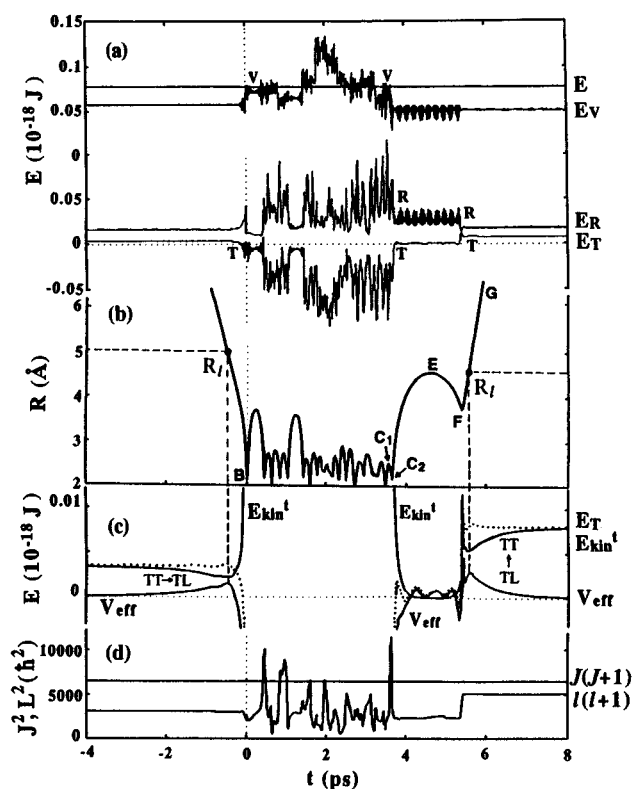


FIG. 5. Time-dependent partitioning of energy and angular momentum of a reactant NO_2 pair in the gas phase at $E=0.078 \times 10^{-18}$ J: (a) total energy E and energy partitioning among E_T , E_R , and E_V , where the upper ordinate is for E and E_V , and the lower one is for E_R and E_T ; (b) interfragment distance R ; (c) E_T (dotted line), E_{kin}^t and V_{eff} (full lines) during the approaching and separating processes of fragments; (d) $J(J+1)$ and $l(l+1)$. The position of the orbital angular momentum dependent effective barrier R_l is shown in the figure.

seen. The TS in the sense of point of no return in this trajectory is hence F ($R=3.75$ Å), which is located at the inner side of the interfragment motion than E in the bound state near threshold. E_T vs R curve in Fig. 6, which is plotted from C_1 to G of the trajectory in Fig. 5(b), clarifies the situation. The concept of the variational RRKM TS at $R_l^\ddagger$, where i denotes the quantum numbers of the conserved modes,¹⁷ fails here microdynamically because the TS point F is at smaller interfragment separation than E in the A state with almost the same vibrational energy. Rotational energy, as well as vibrational energy, is not always available to the reactive mode. These observations suggest that the state of a system is determined by the sign of E_T , which can be determined only by specifying both E_V and E_R . Energy transfers among $T-R-V$ modes at B and C_2 are both centered at ITPs, while RT energy transfer at F occurs near an OTP of the interfragment motion. Orbital angular momentum $l(l+1)$ is nearly conserved before association at B, near the threshold region from C_2 to F, and after dissociation at F [Fig. 5(d)]. As a result, E_{kin}^t becomes minimum at an l -dependent potential barrier R_l of the effective potential V_{eff} , keeping E_T nearly constant. Energy transfer from the interfragment translational mode (TT) to the orbital mode

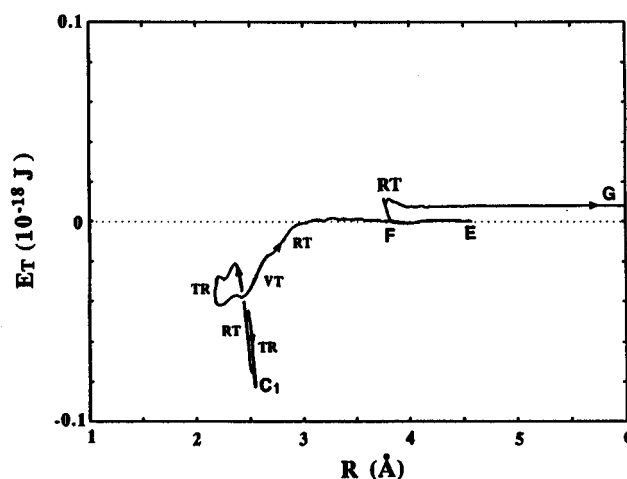


FIG. 6. Reactive mode energy E_T versus interfragment separation R of the dissociation process [from C_1 to G in Fig. 5(b)]. A point of no return of the trajectory is at F where the T mode gets enough energy to dissociate by RT energy transfer. Energy transfer responsible for each energy fluctuation in E_T is shown in the figure.

(TL) upon association and the one from TL to TT upon dissociation are clearly observed around R_l 's [Fig. 5(c)]. E_T is kept constant during the energy transfer processes, and there is no indication of energy fluctuation among $T-R-V$ modes around R_l .

Figure 7 shows a trajectory which includes a chaotic interfragment motion near the dissociation threshold. This trajectory is initiated from the same configuration as in Fig. 2 except it is excited more ($R^{\text{ex}}=1.4355$ Å, $E=0.134 \times 10^{-18}$ J). The fragments associate at an ITP B by TR energy transfer and dissociate at another ITP H by VT and RT energy transfer. Three large-amplitude interfragment motions during CD, DE, and FG in the figure are characterized by $E_T \approx 0$. The TS of the association and dissociation reactions of this trajectory is at B and H, respectively, where E_T changes sign by intermode energy transfer. These points are at smaller interfragment separation than points P, Q, and R, which are close to $E_T=0$ but still in the bound side. Excess energy in the V and R modes is found not available during 9 ps of large-amplitude motion near threshold.

Figure 8 shows a trajectory initiated from the same configuration as in Fig. 5 except it is excited less ($R^{\text{ex}}=1.5200$ Å, $E=0.061 \times 10^{-18}$ J). A pair of fragments collides and associates at B by TV and TR energy transfer, and dissociate at C by VT and RT energy transfer. It is noted that a deceleration followed by acceleration of interfragment separation is clearly observed in the dissociation process [Fig. 8(b) and E_{kin}^t of Fig. 8(c)]. This type of deceleration and acceleration occur before and after passing through an orbital angular momentum dependent barrier R_l .

Figure 9 shows an example of dissociation reactions that proceed, keeping E_T , E_R , and E_V conserved, where $E=0.116 \times 10^{-18}$ J. The trajectory is activated at B by VT energy transfer, gets positive E_T at C by RT energy transfer, and is reflected at E by a high barrier of V_{eff} [$l(l+1) \approx 2600$]. Then an abrupt energy transfer from TL to TT

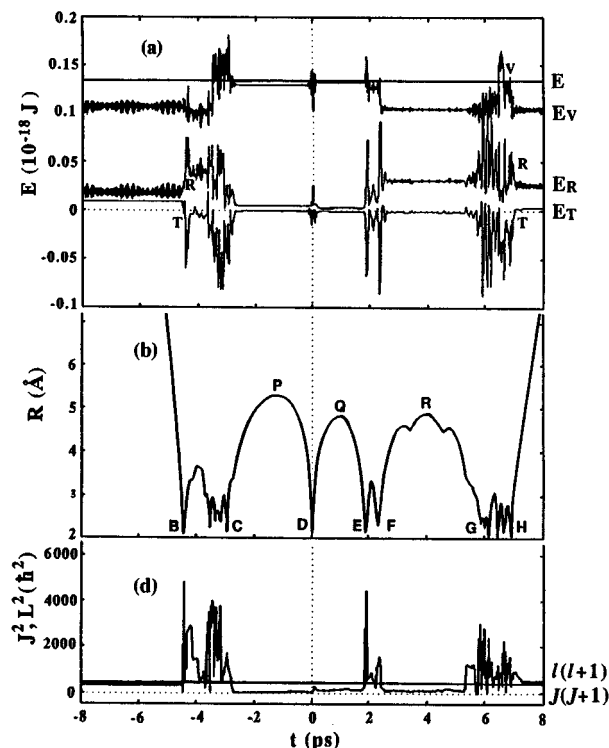


FIG. 7. Time-dependent partitioning of energy and angular momentum of a chaotic interfragment motion near the dissociation threshold at $E=0.134 \times 10^{-18}$ J: (a) total energy E and energy partitioning among E_T , E_R , and E_V ; (b) interfragment distance R ; (c) $J(J+1)$ and $l(l+1)$.

occurs in the midway F, and the conserved E_T becomes energetic enough to pass over a lowered effective barrier [$l(l+1) \approx 530$], leading to dissociation.

From the simulations initiated from the same configuration of a reactant NO_2 pair, except for various R^{ex} values, the lifetime τ of the associated state is obtained and is plotted as a function of the reactant's total energy E in Fig. 10. It is found that $\tau \rightarrow \infty$ at low E and $\tau \rightarrow 0$ at high E because a system is always associated at low energy and dissociates directly at high energy. However, τ of each trajectory is found not to be a monotonically decreasing function of E , but shows two orders of magnitude variations within a narrow energy range, even if the initial condition is the same except R^{ex} , the NN distance at the ITP of interfragment motion. The lifetime of a trajectory is determined by whether a reactive intermode energy transfer occurs or not at each ITP. This may be influenced by many incidental factors, such as the amplitudes and phases of the reactive, conserved, and transitional modes. Only a slight change of one of these, the amplitude of the reactive mode in the present case, is found to change the lifetime drastically. However, the striking fluctuations of τ as shown in Fig. 10 will not be observed experimentally in medium size or large molecules because it is difficult to excite the system to a selected initial configuration. We observe only the averaged quantities that depend on experimental conditions.

The fate of a trajectory, whether it will dissociate or remain associated, is found to be determined not when the

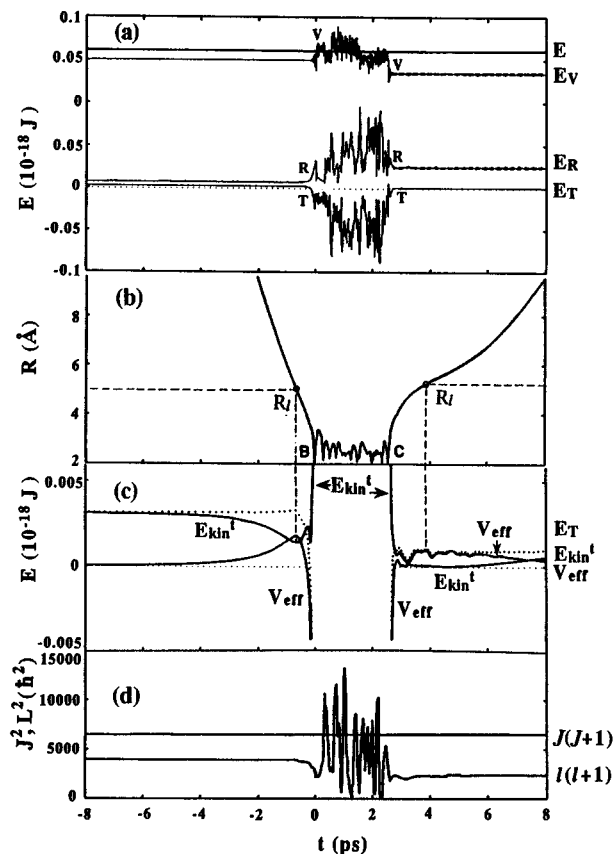


FIG. 8. Time-dependent partitioning of energy and angular momentum of a reactant NO_2 pair in the gas phase at $E=0.061 \times 10^{-18}$ J: (a) total energy E and energy partitioning among E_T , E_R , and E_V , where the upper ordinate is for E and E_V , and the lower one is for E_R and E_T ; (b) interfragment distance R ; (c) E_T (dotted line), E_{kin}^t and V_{eff} (full lines) during the approaching and separating processes of fragments; (d) $J(J+1)$ and $l(l+1)$.

reaction coordinate reaches a special value $R=R^\ddagger$ but when the reactive T mode acquires enough energy to surmount the dissociation threshold. The TS in the sense of point of no return in no-barrier reactions should therefore be defined in terms of a phase space surface $E_T=0$, or more strictly $E_T=V_{\text{eff}}(R_l)$, which is slightly positive. It is noted that the orbital angular momentum l and hence the effective barrier height $V_{\text{eff}}(R_l)$ is determined in the final stage of the dissociative energy transfer, a little later than the main VT transition. We have found from the simulations of activated N_2O_4 in the gas phase the following facts.

- (i) Total energy E and total angular momentum J is conserved in the absence of collisions.
- (ii) $E_T > 0$ in the D state and $E_T < 0$ in the A state.
- (iii) When a pair of fragments approaches or departs at constant E_T in the D state, E_V and E_R of fragments are also kept constant, which shows that the vibrational and rotational motions are those of free molecules irrespective of the interfragment separation R .
- (iv) Interfragment motion is oscillatory in the A state. The T and R modes are strongly coupled, and energy fluctuation of half of the period of interfragment oscillation is

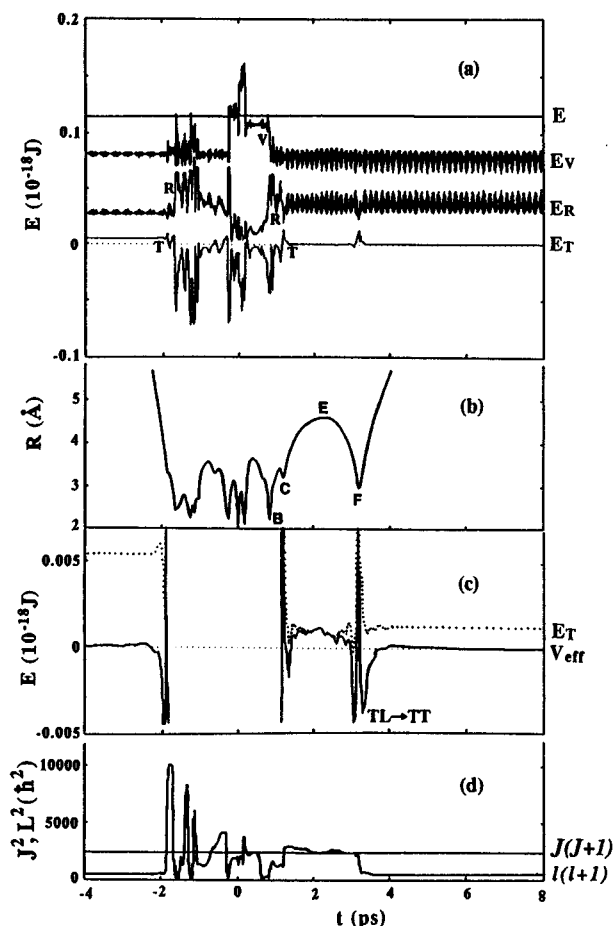


FIG. 9. Time-dependent partitioning of energy and angular momenta of a dissociation reaction, which proceeds keeping E_T , E_R , and E_V conserved at $E = 0.116 \times 10^{-18}$ J: (a) total energy E and energy partitioning among E_T , E_R , and E_V ; (b) interfragment distance R ; (c) E_T (dotted line) and V_{eff} (full line) during the approaching and separating processes of fragments; (d) $J(J+1)$ and $l(l+1)$.

observed in E_T and E_R , while marked vibrational energy fluctuation is less often and occurs at an ITP of interfragment motion by the short range repulsive interaction.

(v) Orbital angular momentum $l(l+1)$ is conserved in the D state and fluctuates periodically in the A state. Conservation of l gives rise to an l -dependent potential barrier at $R = R_l$ in the effective potential V_{eff} when l is large. When fragments approach with constant l , energy transfers from TT to TL mode upon association, keeping their sum E_T positive constant. Energy transfers from TL to TT during separation of fragments.

(vi) Association occurs when a pair of fragments approaches over the effective barrier [$E_T > V_{\text{eff}}(R_l)$] and when E_T changes sign from positive to negative by TV and/or TR followed by RT energy transfer, and dissociation into fragments occurs when E_T changes sign from negative to positive enough to pass over the effective barrier [$E_T > V_{\text{eff}}(R_l)$] by VT and/or TR followed by RT energy transfer. These energy transfers that lead to D/A reactions are centered at ITPs of the interfragment motion.

(vii) Least occasionally, the D/A reactions occur by

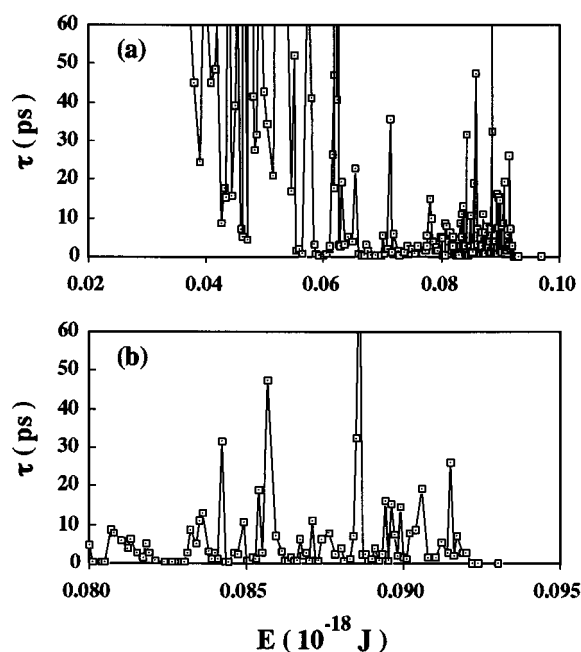


FIG. 10. (a) Lifetime τ of the associated state as a function of total energy $E = 0.02$ – 0.10×10^{-18} J from the simulation of a reactant NO_2 pair for various R^{ex} values; (b) τ for a narrow energy range $E = 0.08$ – 0.095×10^{-18} J. τ is found not to be a monotonically decreasing function of E , but shows striking variations within a narrow energy range.

RT/TR energy transfers, respectively, near OTPs.

These observations (i)–(vii) in the N_2O_4 reaction can be generalized to other no-barrier reactions except the following facts: TV , VT , and TR followed by RT energy transfers are dominantly centered at ITPs of interfragment motion in N_2O_4 . This may not be true in other reactions because the parts of interfragment motion where intermode energy transfers occur depend on the details of intra- and interfragment potentials.

B. Dynamical model for reactive energy transfer

We can propose a common scheme of energy and angular momentum transfer in no-barrier fragmentation reactions in the gas phase, which is shown in Fig. 11. Time intervals during which a dynamical variable is conserved are shown by solid lines. Total energy E and total angular momentum $J(J+1)$ are conserved over all processes of reaction. E_T is kept positive in the direct scattering DD process, and E_T changes sign (+, −, +) corresponding to the DAD process. When a pair of fragments approaches, energy transfers from the TT to the TL mode with constant E_T and constant l . The pair is reflected by the effective barrier to association if $E_T \leq V_{\text{eff}}(R_l)$, and the pair passes over the barrier and collides at an ITP of the Morse+LJ potential if $E_T > V_{\text{eff}}(R_l)$. Fragments associate when E_T changes sign from positive to negative by TV energy transfer and/or TR followed by RT energy transfer, which are centered at the ITP. In rare cases fragments associate by TR energy transfer near an OTP.

Energy transfer among T – R – V modes continues during the pair is associated, which is an energy redistribution

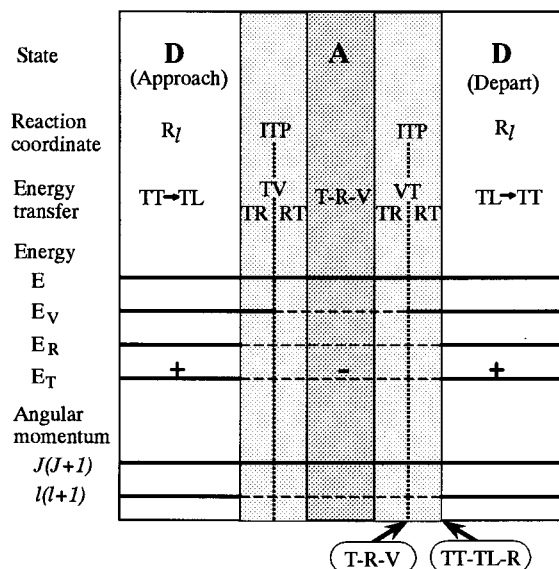


FIG. 11. Common scheme of reactive energy and angular momentum transfer in the association and dissociation reactions in the gas phase. Time intervals during which a dynamical variable is conserved are shown by solid lines. First the vibrational state of products is determined near an ITP by energy redistribution among $T-R-V$ modes, and second product rotational and translational state distribution is determined during separation by energy redistribution among $TT-TL-R$ modes.

process among vibrational degrees of freedom of the parent molecule. The molecule then dissociates when the VT process and/or TR followed by the RT process change the sign of E_T from negative to positive enough to pass over the l -dependent barrier. The VT process occurs by an energy jump at an ITP, and TR followed by the RT process is centered at the ITP and terminates farther toward products. In rare cases a molecule dissociates by RT energy transfer near an OTP. Then the fragments separate transferring energy from TL to TT , while conserving E_T and l . The barrier energy appears exclusively as relative translational energy in the separated products. It is noted that first the vibrational state of products is determined near an ITP by energy redistribution among $T-R-V$ modes, which is most probably among parent vibrational modes excluding parent rotation, and second the product rotational and translational state is determined during separation by energy redistribution among the reactive and transitional modes as well as parent rotation. The second process is among $TT-TL-R$ in the words of a loosely bound pair.

C. Comparison with statistical theories

Let us examine here various TSTs and additional dynamical assumptions proposed to predict PSD from the dynamical point of view. As we have seen in the last section, the fate of a trajectory is determined, not when the reaction coordinate reaches a special value $R=R^\ddagger$ but when the reactive T mode acquires enough energy to surmount the dissociation threshold. The TS in no-barrier reactions such as the present N_2O_4 reaction should be defined in terms of a phase space surface $E_T=0$, or more strictly $E_T=V_{\text{eff}}(R_l)$, which is

slightly positive. This conclusion seems to be applicable to a large part of no-barrier fragmentation reactions in which the reactive T mode is a good mode and interfragment motion is represented by the T mode energy E_T .

In PST, the TS is placed at the barrier R_l of the effective potential V_{eff} , and the reactive mode energy E_T must exceed the l -dependent barrier in order for a reaction to occur. The energy condition for the D/A reactions from present MD simulations is completely in accord with the one of PST. The observed slowly increasing dissociation rate with energy in photofragmentation than predicted by PST may be explained by developing models that explicitly incorporate the rate of intermode energy transfer.

The transition that crosses the TS surface $E_T=V_{\text{eff}}(R_l)$ is found to occur by three steps: first, energy redistribution among $T-R-V$ modes at an ITP of interfragment motion, where E_T becomes positive and the product vibrational distribution is determined; second, energy redistribution among $TT-TL-R$ modes, where l is determined, E_T becomes larger than $V_{\text{eff}}(R_l)$, and the product rotational distribution is determined; and third, the fragments separate transferring energy from TL to TT with constant E_T and l . This scheme of reactive energy transfer justifies the agreement of the SSE method, the revised version of PST, with the experimental nascent distributions from photofragmentation of CH_2CO and $NCNO$ at excess energies above the vibrational threshold.^{15,16}

Variational RRKM theory for unimolecular reactions and additional dynamical assumptions introduced to calculate PSD^{11,17} have been considered to be the only available model that is consistent with all the experimental data of photofragmentation of $NCNO$ and CH_2CO .^{10,12,14} The theory that assumes vibrational adiabaticity predicts rates on each VA surface i by placing the TS at the value $R_i^\ddagger$ that minimizes $N_{iEJ}^\ddagger$. Many contradictions coming from defining the TS in coordinate space alone have been pointed out in the last section. The motion along the reaction coordinate is often as rapid as, or more rapid than the orientational motion for the transitional modes after crossing the TS surface $E_T=0$, and the number of accessible states is more close to the free rotor asymptote considered in PST than the one calculated by Eq. (5) at the variationally determined TS.

In the VA version of variational RRKM, excess energy $E-E_i^\ddagger$ in Eq. (5) is partitioned to the T and R modes by assuming that the rotational energy is always available to the reactive mode for surmounting the potential barrier. However, excess energy in the V and R modes was not available during 9 ps of large-amplitude motion near the threshold in Fig. 7. Rotational energy, as well as vibrational energy, was found not available to the reactive mode, and the assumption of adiabatic vibrational and statistical rotational motions used in variational RRKM theory is not justified microdynamically. The variational RRKM theory for the no-barrier dissociation reaction is found not valid because $N_{iEJ}^\ddagger$ contains the number of states with $E_T<0$, which are sure to recross unless RT energy transfer occurs just after passing the assumed TS.

V. CONCLUDING REMARKS

Thermal dissociation and association reactions on the ground state no-barrier potential energy surface have been studied by classical MD simulations of $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$. A phase space surface $E_T = V_{\text{eff}}(R_f) \geq 0$ was identified as the TS that has a point of no return property to product. A unified picture of reactive energy and angular momentum transfer for the fragmentation reaction into polyatomic molecules is presented in terms of an interfragment $T(=TT+TL)$ mode, and V and R modes of fragments.

Various transition state theories that place the TS in configuration space, and more generally in phase space, were examined from the dynamical point of view. In no-barrier reactions near the threshold, the number of accessible states at the PST TS is small and the rate of dissociation is very low. Because energy is randomized faster than dissociation in this case, the rate can be determined statistically as in Eq. (2). As energy is increased above the threshold for reaction, the dissociation rate increases and may become comparable to the intrareactant energy transfer rate. In this energy range, energy randomization among degrees of freedom may influence the rate, and the rate calculated statistically will give only an upper limit.

Both variational RRKM and SACM gave better fit to the observed slowly increasing dissociation rate with energy by decreasing the number of accessible states at the TS in configuration space. However, the observed dissociation rates may be explained by developing models that explicitly incorporate the rate of intrareactant energy redistribution. In a forthcoming paper, the fragmentation rate for an arbitrary TS surface will be derived. The rate will be strongly dependent on the definition of the TS, and the rate for the TS surface $E_T=0$ in phase space will contain the contribution of the rate of intermode energy redistribution. In this paper, we confined our interest to the reaction processes of N_2O_4 activated above the dissociation threshold without the effect of intermolecular collisions. The next subject will be the reaction dynamics in liquid phase where the reactant's T mode interchange energy and angular momentum with solvent modes as well as reactant's R and V modes.

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